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Course 201

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Research Topic: HEXAGUARD: An IoT-Based Search and Rescue Rover for Hazard Monitoring and Emergency Response

Abstract

Search and rescue operations often take place in dangerous environments such as fires, building collapses, industrial accidents, and other emergencies. Smoke, toxic gases, high temperatures, poor visibility, and unstable structures can put both victims and rescue workers at risk. In such situations, rescuers may need to enter dangerous areas to understand the environment, locate victims, and provide assistance. This creates a need for systems that can collect useful information from hazardous areas while keeping rescue workers at a safer distance.

Robotics and IoT technologies can help address this problem by allowing remotely controlled systems to monitor and inspect difficult environments. However, many existing rescue robots focus on a limited number of functions, such as environmental monitoring, gas detection, live video, or remote movement. This research proposes HEXAGUARD: An IoT-Based Search and Rescue Rover for Hazard Monitoring and Emergency Response, a remotely operated rover designed to combine several rescue-related functions in one platform. The proposed system will integrate hazard monitoring, live visual observation, remote mobility, robotic object handling, and basic fire suppression.

The proposed system will be tested in different conditions to examine its performance in hazard monitoring, mobility, object handling, and basic emergency response. The research will also identify the system's limitations and examine its potential as a supporting tool for search and rescue operations.

Introduction

Search and rescue (SAR) operations have become more difficult because of fires, building collapses, industrial accidents, earthquakes, and other disasters [1], [2]. These situations can create dangerous environments. There can be smoke, toxic gases, high temperatures, damaged buildings, and poor visibility [1], [3]. As a result, rescue workers face serious risks while trying to save lives [4].

When fires or building collapses occur, rescue teams may be unable to enter the area because of heat, smoke, toxic gases, or the danger of further collapse. Recent advances in IoT and robotics have made rescue robots useful for disaster response [2], [8]. Many systems can provide live video, *observe environmental conditions*, and collect sensor data from dangerous areas without directly exposing rescue workers [1], [9]. However, many existing systems still focus on limited functions such as gas detection, live video, or remote control. Integrating functions such as multi-gas monitoring, object manipulation, and fire suppression remains less common [6], [7], [10].

Therefore, this research proposes HEXAGUARD: An IoT-Based Search and Rescue Rover for Hazard Monitoring and Emergency Response. The research will focus on developing a remotely operated rover that combines multi-gas detection, hazard monitoring, live visual observation, metal detection, dual communication, mobility, robotic object handling, and basic fire suppression within a single platform.

2. Existing Literature

Search and rescue robots have become an important research topic because they can reduce risks faced by rescue teams during disasters [11], [12]. With the growth of IoT, embedded systems, sensors, and wireless communication, researchers have developed rescue robots for environmental monitoring, victim detection, surveillance, and remote operation [6], [7], [13].

12 Sharmin et al. [6] proposed a low-cost urban search and rescue robot for developing countries using gas and environmental sensors, a thermal camera, HC-12 communication, and a web dashboard. The robot also uses a four-wheel-drive chassis for movement on uneven ground.

Saha et al. [7] developed Hazard Hunter, a low-cost ESP32-based rescue robot using MQ-2, DHT11, and ESP32-CAM for gas monitoring and live video. It is controlled through a Wi-Fi web interface.

Pragatheeswari et al. [13] presented an IoT-based smart rover using ESP32 and ESP32-CAM with DHT11, MQ-2, flame, ultrasonic, and vibration sensors. The system uses 15 Blynk for real-time monitoring and alerts.

22 Raju et al. [14] introduced a six-wheel rescue and surveillance robot using Arduino UNO and Raspberry Pi. It includes a robotic arm, webcam, and mobile application for remote Wi-Fi control.

After reviewing these studies, several common points can be seen. Most rescue robots use Wi-Fi or mobile applications for both rover control and monitoring [6], [7], [13], [14].

Most systems also use only a few sensors, mainly for gas detection, temperature, humidity, flame, or distance measurement [6], [7], [13]. Some robots include a robotic arm [14], while others focus on victim detection [6] or environmental monitoring [13]. However, there are still some limitations.

Most existing robots mainly monitor the environment and send live video, but only a few can actively help during rescue missions. Features like fire suppression, multi-gas monitoring, and modular hardware expansion are still uncommon [6], [7], [13], [14]. In

addition, many systems are designed mainly for indoor testing, with limited discussion about working on grass, debris, uneven ground, or during light rain [7], [13], [14].

Because of these limitations, there is still a need for a rescue robot that combines easy control, real-time monitoring, multi-gas detection, robotic manipulation, fire suppression, and a modular design on one platform.

3. Research Question

Central Research Question:

- How can an IoT-based search and rescue rover be developed to integrate multiple hazard detection, communication, monitoring, manipulation, and emergency response functions within a single platform for use in hazardous environments?
- How effectively can the rover detect multiple hazardous gases and metal objects?
- How reliably can the dual RF and Wi-Fi systems provide remote control and real-time monitoring?
- How effectively can the rover perform basic object-handling tasks using its robotic arm and water-based fire suppression?
- How well can the six-wheel rover operate on different surfaces?

4. Methodology

The research will be conducted by designing, developing, integrating, and testing the HEXAGUARD search and rescue rover. The methodology will include the overall system design, communication and control, sensor allocation, rover locomotion and actuation, operational workflow, and experimental testing. The different systems will be integrated into one remotely operated platform so that the rover can collect information from hazardous environments and perform basic rescue-support tasks.

4.1 System Design and Control

HEXAGUARD will use a six-wheel differential-drive chassis with a stainless-steel body, two ESP32 controllers, multiple gas sensors, two IP cameras with two-way audio, a 6-DOF robotic arm, metal detection, LED lights, and a water-based fire suppression system. The two ESP32 controllers will divide sensing and control tasks to make the system more modular.

The rover will use hybrid RF and Wi-Fi communication. A FlySky FS-i6X transmitter and X6B receiver will provide rover movement and actuator control, while Wi-Fi will

connect the ESP32 controllers to a web-based monitoring dashboard for sensor readings, device status, and live camera feeds.

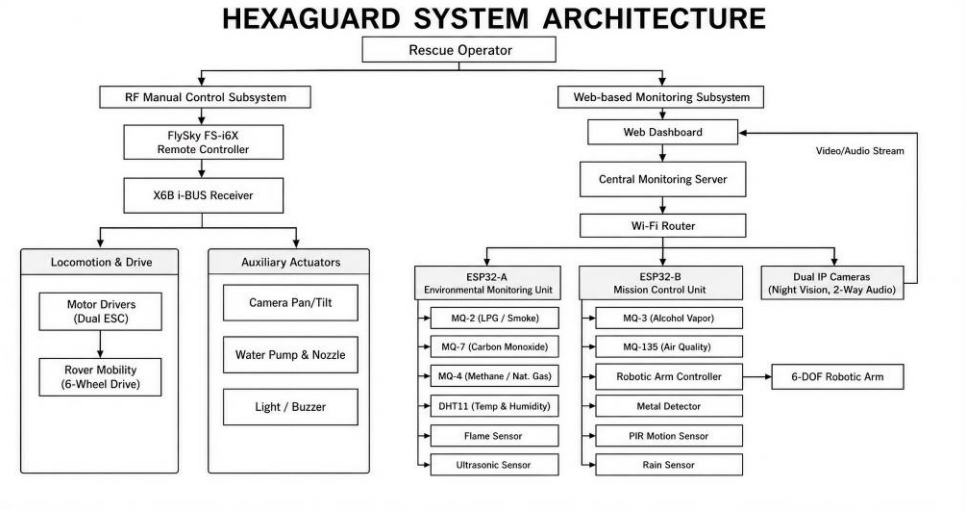


Fig. 1. Overall System Architecture of HEXAGUARD.

ESP32-A will handle MQ-2, MQ-7, MQ-4, DHT11, flame, and HC-SR04 sensors. ESP32-B will handle MQ-3, MQ-135, metal detection, PIR, rain sensing, and the 6-DOF robotic arm. This distribution will reduce processing and wiring limitations and allow easier expansion.

4.2 Locomotion, Actuation, and Operation

The rover will use a six-wheel differential-drive system controlled by two electronic speed controllers, with each controller operating the motors on one side. Its stainless-steel body will help protect the electronics from dust and light rain. A pan-tilt mechanism will allow the camera to change direction, while the robotic arm will be used for basic object-handling tasks. A relay-controlled water pump and servo-controlled nozzle will be used to suppress small fires in a localized area.

During operation, the rover will be controlled through RF while the ESP32 controllers collect sensor data and send it through Wi-Fi to the monitoring dashboard. When smoke, gas, carbon monoxide, methane, or flame reaches a predefined threshold, the system will alert the operator. The operator will then decide whether to reposition the rover, operate the arm, activate the pump, or continue observation. The system will remain under human supervision.

4.3 Experimental Testing and Data Analysis

The system will be tested for hazard detection, metal detection, communication, mobility, object manipulation, and basic fire suppression. Gas sensors will be tested through

repeated detection trials, while metal detection will be tested using metallic objects. RF and Wi-Fi communication will be observed during operation.

The rover will be tested on grass, uneven ground, and mixed terrain. The robotic arm will perform repeated pick-and-place tasks using lightweight objects. The fire suppression system will be tested using a small controlled flame. Results will be analyzed using detection and task-success rates by comparing successful trials with total trials. The study will also record practical limitations related to terrain, object weight, communication, and fire-suppression capacity.

5. Expected Outcomes

The research is expected to result in a functional HEXAGUARD rover that can monitor hazards in real time, be controlled remotely, provide live visual information, move across different surfaces, handle basic objects, and help suppress small fires. Through testing, the study will examine how well these functions work and identify the strengths and limitations of combining them in one remotely operated platform for search and rescue support.

6. Limitations

The rover will still need a person to control it during operation, so it will not work fully on its own. The robotic arm may have difficulty handling heavier objects, and the fire suppression system will mainly be suitable for small controlled fires because of its limited water capacity and pump capability. Sensor and communication performance may also change depending on the environment and operating conditions.

7. Conclusion

This research proposes HEXAGUARD, an IoT-based search and rescue rover designed to detect hazards, provide live monitoring, move across different surfaces, handle basic objects, and support small-fire suppression through hybrid RF and Wi-Fi communication. The system will be tested to see how well these functions work and to understand its practical strengths and limitations.

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The main goal of this research is to examine whether combining these functions in a single remotely operated platform can provide useful support during search and rescue operations in hazardous environments. By keeping the operator at a safer distance while allowing the rover to collect information and perform basic tasks, the study will explore the potential of robotics and IoT as practical tools for supporting rescue teams.

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